

Report.kg – Bug Reporting & Issue Management Platform

1. Introduction

Report.kg is a web-based bug reporting and issue management platform designed for organizations and institutions that require a structured system to manage technical issues, operational complaints, and department-based workflows. The platform enables users to report issues efficiently while allowing departments to verify, process, and resolve them through a centralized dashboard.

The system is designed as a Progressive Web App (PWA), allowing access from desktop and mobile devices without requiring a dedicated mobile application. Report.kg aims to improve communication between users and departments while ensuring transparency, faster issue resolution, and proper activity tracking.

2. Objectives

- Create a centralized platform for issue and bug reporting.
 - Support multiple departments within a single system.
 - Improve issue tracking and resolution workflows.
 - Provide secure document and file handling.
 - Generate reports and analytics for management.
 - Enable mobile-friendly reporting through PWA support.
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3. Core Features

Multi-Department Support

The platform allows multiple departments to operate independently within the same system. Each department can manage its own reports, staff, and workflows.

Role-Based Access Control

Different user roles are supported, including Admin, Department Staff, and End Users. Access permissions are controlled based on user responsibilities.

Structured Bug Reporting

Users can submit detailed issue reports with categories, descriptions, priority levels, and supporting files.

Dashboard Analytics

Interactive dashboards provide insights into issue statistics, pending tasks, resolved reports, and departmental performance.

Issue Tracking System

Every issue is assigned a unique tracking ID, enabling users and departments to monitor the progress of reports in real time.

PWA Support for Mobile

The platform functions as a Progressive Web App (PWA), ensuring compatibility across mobile and desktop devices with responsive design and installable support.

Report Generation

Departments and administrators can generate downloadable reports for audits, monitoring, and management analysis.

4. Supported File Formats

The platform supports secure uploads of multiple document and media formats, including:

- PDF
- CSV
- XLS/XLSX
- JPG
- PNG
- ZIP Files

These uploads help users attach screenshots, logs, spreadsheets, and supporting documents to issue reports.

5. Workflow

The operational workflow of Report.kg follows a structured process:

1. User Login
2. Bug/Issue Report Submission
3. Validation of Fields and Documents
4. Submission to Relevant Department
5. Department Verification

- 6. Processing and Resolution
 - 7. Status Updates
 - 8. Closure of Issue
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6. Technology Stack

Frontend

- React.js / Next.js

Backend

- Node.js / NestJS

Database

- PostgreSQL

Storage

- Cloudflare R2

Cloudflare R2 provides scalable cloud storage for uploaded files and documents.

7. Security Features

Security is a core component of Report.kg. The platform includes:

- Role-Based Authentication
- JWT Authorization
- Secure File Upload Validation
- Activity Logging
- Audit Trails